

Supplementary Material

The Divergence Hypothesis: Unmasking Lexical Interference and Label Bias in Mental Health NLP

Anonymous ACL BioNLP 2026 Submission

This document contains the supplementary appendices for the main paper. All tables, formulations, and diagnostics referenced in the paper are provided below.

Appendices

A Mathematical Formulations Summary

This appendix collects the novel mathematical contributions of this work for reference.

A.1 Degree of Divergence (DoD)

$$\text{DoD} = (M_h(\text{BC}) - M_h(\text{A})) - (M_a(\text{BC}) - M_a(\text{A})) \quad (1)$$

Adaptation of the difference-in-differences (DiD) causal framework from econometrics to quantify annotation regime bias in NLP. Bootstrap inference ($N=10,000$) over all 12,906 instances.

A.2 Adaptive Length-Smoothing (Channel B)

$$\lambda = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \times \min(L/L_{\text{ref}}, 1.0) \quad (2)$$

Piecewise linear smoothing with $\lambda_{\min}=3.0$, $\lambda_{\max}=20.0$, $L_{\text{ref}}=50.0$. Ensures short texts retain sparse POS signals while long texts are regularized.

A.3 Length-Normalized Log-Odds Mass (Channel B)

$$B_{\text{pos_mass}} = \frac{\log(1 + M_{\text{pos}})}{\log(1 + L_{\text{eff}})}, \quad \text{where } L_{\text{eff}} = L + \lambda \quad (3)$$

Logarithmic dampening in the denominator prevents length-driven inflation of mass values in long Reddit posts. The remaining Channel B features are: $B_{\text{polarity}} = M_{\text{pos}}/(M_{\text{pos}} + M_{\text{neg}} + \varepsilon)$, $B_{\text{load}} = \log_e(1 + M_{\text{pos}} + M_{\text{neg}})/\max(\log_e(1 + \text{cnt}), 1)$, $B_{\text{abs}} = |\{t : t \in \mathcal{A}\}|/L_{\text{eff}}$ where $\mathcal{A}$ is a closed absolutist-word list. Note: B_{abs} is the sole Channel B feature that accesses lexical tokens.

A.4 Structural Negation Balance (Channel C)

$$\text{NegScopeBalance} = \frac{R(\text{Neg} \rightarrow \text{Verb}) - R(\text{Neg} \rightarrow \text{Adj})}{\text{TotalNegEvents} + \varepsilon} \quad (4)$$

Operationalizes the clinical distinction between behavioral helplessness and negative self-evaluation as a computable structural feature.

A.5 Length-Robust Squashing (Channel C)

$$c_{\text{out}} = \tanh(2 \cdot c) \quad (5)$$

Pushes strong signals to saturation (± 1), converting continuous features to soft binary triggers resistant to cross-platform distributional shift.

A.6 Length-Independent Lexical Diversity (Yule's I)

$$I = \frac{M_1^2}{M_2 - M_1} \quad (6)$$

Where M_1 is the number of unique tokens (vocabulary size), and $M_2 = \sum f_i^2$ (the sum of squared token frequencies).

Rationale for Cross-Platform Transfer: Traditional metrics for lexical diversity, such as Type-Token Ratio ($\text{TTR} = V/N$), mathematically decay as text length increases, creating a severe structural bias when transferring models from lengthy Reddit posts to short Twitter posts. We introduce Yule's I characteristic into Channel C because it treats vocabulary distribution as a probabilistic urn model, rendering the diversity score asymptotically independent of document length. This allows TSS to consistently measure "rumination" and "lexical repetitiveness" across wildly different platform constraints without introducing domain gaps.

B Qualitative Conflict Zone Analysis

To contextualize the quantitative performance gaps and illustrate the Lexical Interference Paradox, we conducted an instance-level qualitative analysis on the “Conflict Zone”—samples extracted via our error-analysis pipeline where the lexical baseline (Channel A / Auto-labels) strongly disagreed with the TSS structural probes (Channels B and C).

Table B.1 illustrates how purely lexical systems fall prey to Semantic Hallucination (over-indexing on keywords) and Context Contagion, while missing Implicit Stress (structural fragmentation lacking explicit diagnostic vocabulary).

Table B.1: Instance-Level Qualitative Analysis of Lexical vs. Structural Inference.

Case Type	Real Text Sample	GT	Lexical (A)	TSS (B/C)	Rationale
1. Semantic Hallucination (Keyword Spotting)	“call us telephone receiver at +971 4 324 3244 to consult a medical professional. EyeHealth MentalHealth Ophthalmologist Stress DryEye EyeTwitch” (Source: reddit_combi / Twitter)	0	1 (Stress)	0	The lexical baseline is fooled by diagnostic hashtags (#MentalHealth, #Stress). TSS rejects this as spam/advertisement: imperative syntax (“call us”), no first-person pronouns, no cognitive fragmentation.
2. Implicit Stress (Missing Vocabulary)	“I am on the edge of committing and I have a few hours remaining to decide whether I end my life.” (Source: reddit_combi / dreddit)	1	0 (Non-Stress)	1	The lexical channel misses severe distress because the user avoids explicit keywords. TSS captures structural urgency: dense first-person pronouns, absolutist cognitive state (“edge”, “end”), specific POS action sequences.
3. Boundary Collapse (Context Contagion)	“I man the front desk and my title is HR Customer Service Representative... The rest is dealing with people who walk in and need help with benefits, retirement, etc.” (Source: dreddit_test)	0	1 (Stress)	0	A fires on struggle-associated words (“need help”, “dealing with people”). TSS identifies objective, descriptive narrative with stable declarative syntax and zero structural anxiety markers.
4. Semantic Hallucination (Resilience Coaching)	“Resilience is not a trait that you are either born with or without. It’s a set of behaviors, thoughts, and actions that can be learned and developed. resilience mentalhealth mentalstrength lifelessons” (Source: twitter_gold)	0	1 (Stress)	0	Psychoeducational content tagged with #mentalhealth. A keys on topic markers; C detects stable, didactic syntax (generic third-person framing, absence of first-person fragmentation) and correctly rejects.
5. Semantic Hallucination (Counseling Promo)	“Sometimes love is complicated. Let’s talk it out. mentalhealth Couplescounseling marriagecounseling mentalhealthawareness love health...” (Source: twitter_gold)	0	1 (Stress)	0	Promotional framing for counseling services. A fires on the dense mental-health lexicon; C identifies marketing register: imperative invitations (“Let’s talk”), no negation scope, no rhythm fragmentation.
6. Semantic Hallucination (Product Ad)	“New Product Alert! Silicone Suction Snapper Use code SOCIAL10 for 10% off! Sensory SPD Stim Autism Stress MentalHealth HappyHandsToys” (Source: twitter)	0	1 (Stress)	0	Commercial advertisement tagged with disability/stress hashtags. A detects keyword clusters; C identifies transactional syntax (imperative, product codes, no personal pronouns) incompatible with distress.
7. Semantic Hallucination (Positive Achievement)	“I have a personal goal of squatting 315 lbs for six reps. Two months ago I could only do 195lbs. Tonight I just did 305!!!! I’m so excited!!!!” (Source: reddit_combi)	0	1 (Stress)	0	Celebratory fitness post. A triggers on stress-associated community context; C detects positive valence, exclamatory rhythm (repeated “!!!!”), and achievement-oriented sentence structure.

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Table B.1 continued from previous page

Case Type	Real Text Sample	GT	Lexical (A)	TSS (B/C)	Rationale
8. Semantic Hallucination (Policy Proposal)	“When moving into their tiny house, they would be given a state I.D. with that tiny house’s address on it as well as a list of strict rules they have to follow...” (Source: dreaddit_test)	0	1 (Stress)	0	Third-person policy discussion about housing for homeless populations. A fires on distress-associated lexicon (“homeless”, “evicted”); C detects stable, expository, non-personal syntax.
9. Implicit Stress (Medical Trauma)	“I had a minor surgery on my neck today... Everything was fine—until I realized they needed to strap me down... I hadn’t been in restraints since the medical [redacted], and I completely dissociated.” (Source: dreaddit_test)	1	0 (Non-Stress)	1	PTSD-triggered dissociation without explicit stress lexicon. A misses it; C captures abrupt narrative rupture (“Everything was fine—until”), first-person recounting with absolutist scope (“completely”), and fragmented rhythm.
10. Implicit Stress (Relational Distress)	“When The one person you thought you could talk to blows you off. tired inconsiderate alone dead hurt” (Source: twitter_gold)	1	0 (Non-Stress)	1	Emotional distress conveyed through colloquial language rather than clinical terms. A misses it; C detects dense negative-affect tokens, fragmented sentence structure, and absolutist framing (“The one person”).
11. Implicit Stress (Masked Depression)	“Like you will feel sad all the time then happy for mere moment before your brain hits you with the ‘damn bruh you were faking it for days’ ” (Source: reddit_combi)	1	0 (Non-Stress)	1	Emotional oscillation and impostor-distress without clinical vocabulary. A misses it; C captures absolutist temporal markers (“all the time”, “for days”), erratic rhythm, and negation patterns.
12. Implicit Stress (Burnout)	“We all have stress at work right? It’s not like I am a doctor or run a billion dollar company. It youth sports.. 60 hours a week (even when I had Covid) I have a second interview for a new job.” (Source: reddit_combi)	1	0 (Non-Stress)	1	Occupational burnout minimized via rhetorical deflection. A misses it because the user downplays distress; C captures fragmented punctuation (“..60 hours”), parenthetical asides, and contradiction between self-minimization and exhaustion markers.
13. Implicit Stress (Financial Crisis)	“Due to recent medical issues of mine (I was hospitalized for [redacted]), we had to pay 1,000 dollars for my hospital bills. We paid our bills... but now we have nothing left.” (Source: dreaddit_test)	1	0 (Non-Stress)	1	Financial distress framed as factual reporting without explicit stress vocabulary. A misses the distress; C captures absolutist finality (“nothing left”), first-person plural hedging, and escalating negation scope.

The expanded case inventory above illustrates systematic patterns in the Lexical Interference Paradox. Cases 1 and 4–8 exemplify *Semantic Hallucination* across diverse registers: commercial advertisements (Case 6), psychoeducational content (Case 4), counseling promotions (Case 5), celebratory posts (Case 7), and policy discussions (Case 8). In each instance, Channel A fires on topical keyword clusters (#MentalHealth, #Stress, distress-associated vocabulary) that correlate with stress in closed communities but serve entirely different communicative functions in these contexts. Channel C correctly rejects these by detecting the absence of structural distress signatures—imperative or transactional syntax, stable declarative rhythm, and missing first-person fragmentation—consistent with the SSI hypothesis (H1): lexical shortcuts succeed when semantic–structural intersection is high but fail when SSI weakens.

Conversely, Cases 2 and 9–13 exemplify *Implicit Stress* across a spectrum of distress expressions: PTSD-triggered dissociation (Case 9), relational distress in colloquial register (Case 10), masked depression with impostor-like self-doubt (Case 11), occupational burnout minimized by rhetorical deflection (Case 12), and financial crisis framed as factual reporting (Case 13). In all instances, the person avoids explicit mental health vocabulary yet exhibits structural distress markers that Channel C captures: absolutist temporal framing (“all the time”, “nothing left”), fragmented rhythm and punctuation, abrupt narrative ruptures, and dense first-person negation patterns. These cases provide direct qualitative evidence for hypothesis H3 (Implicit

Stress vs. Semantic Hallucination): structural probes anchor predictions in rhythm, negation scope, and function-word dynamics, reducing both false-positive and false-negative failure modes.

Workbook statistics. The full error-analysis workbook (*error_analysis_qualitative.xlsx*), available in the anonymized repository, contains four analysis sheets across 800 sampled instances (200 per dataset). The *Shift_Cases_AvsB* sheet documents 682 instances where Channel B outperforms Channel A by correcting A’s errors; these comprise 415 (60.9%) cases where A produces false positives (Semantic Hallucination) and 267 (39.1%) cases where A produces false negatives (Implicit Stress), consistent with the Lexical Interference Paradox failure modes under weakened semantic–structural intersection (SSI). The complementary *Reverse_Shift* sheet documents 667 instances where A corrects B; critically, these split into 319 cases on human-labeled data (Twitter-gold: 200; Dreddit-test: 119) and 348 cases on auto-labeled corpora (Twitter-auto: 200; Reddit-combi: 148). We treat the human-labeled subset as the primary structural failure set: within these 319 gold-supervised reverse shifts, Channel B commits 177 false positives and 142 false negatives against human annotations. Rather than contradicting H1, this B-failure regime is naturally explained by SSI complementarity: lexical Channel A can be selectively reliable when SSI is strong, i.e., when distress is expressed with explicit, domain-typical vocabulary in contexts where the structural cues emphasized by B are less diagnostic, whereas outside that regime A exhibits systematic hallucination and implicit-stress blindness that B is designed to mitigate. The *All_Channels_Compare* sheet provides a full prediction matrix across all seven channel configurations (A, B, C, AB, AC, BC, ABC) for the 800 sampled instances: Channel C alone corrects 253 of Channel A’s errors (195 Semantic Hallucinations + 58 Implicit Stress cases), while the combined BC channel corrects 241. Finally, the *Gold_Disagreement* sheet samples 200 instances for targeted audit of label/decision disagreement patterns, and all instance IDs, per-channel predictions, and qualitative annotations are preserved for reproducibility.

C Additional Statistical Diagnostics

C.1 C.1 SSI Signal-Saturation Test (reddit_combi)

This table reports the paired instance-bootstrap equivalence check underlying the SSI signal-saturation claim in §5.1.

Table 11a: SSI signal-saturation equivalence test (paired instance bootstrap).

Setting	Comparison	Δ Macro-F1 (BC – C)	p (two-sided)
reddit_combi (Auto)	TSS-BC vs TSS-C	+0.0080	0.259

C.2 C.2 Representation Efficiency (Twitter-gold)

This table complements McNemar decision-level tests by reporting paired bootstrap inference over Macro-F1 differences.

Table 11b: Representation efficiency and statistical equivalence (Twitter-gold; paired bootstrap).

Comparison	Δ Macro-F1	95% CI	p
TSS-C – MentalBERT (Tw-gold)	−0.0141	[−0.0367, +0.0082]	0.108

D Detailed Results Tables

Table 5a: Paired baseline comparisons (Δ F1 and paired bootstrap CI).

Pair	Δ F1	95% CI
TSS-C vs MentalBERT (twitter_gold)	−0.0141	[−0.0367, +0.0082]
TSS-C vs LLaMA-3-8B (3-shot) (twitter_gold)	−0.0991	[−0.1206, −0.0775]
TSS-C vs MentalBERT (twitter)	+0.0858	[+0.0732, +0.0985]
TSS-C vs LLaMA-3-8B (3-shot) (twitter)	−0.0183	[−0.0321, −0.0040]
TSS-BC vs MentalBERT (twitter_gold)	−0.0248	[−0.0451, −0.0040]
TSS-BC vs LLaMA-3-8B (3-shot) (twitter_gold)	−0.1099	[−0.1317, −0.0880]
TSS-BC vs MentalBERT (twitter)	+0.0372	[+0.0266, +0.0480]
TSS-BC vs LLaMA-3-8B (3-shot) (twitter)	−0.0668	[−0.0800, −0.0537]

Table 5b: Paired baseline comparisons (McNemar p and d_z).

Pair	p_{McN}	d_z
TSS-C vs MentalBERT (twitter_gold)	4.49e-05	-1.47
TSS-C vs LLaMA-3-8B (3-shot) (twitter_gold)	2.55e-15	-11.40
TSS-C vs MentalBERT (twitter)	4.49e-13	+14.77
TSS-C vs LLaMA-3-8B (3-shot) (twitter)	5.15e-04	-2.83
TSS-BC vs MentalBERT (twitter_gold)	1.81e-04	-2.54
TSS-BC vs LLaMA-3-8B (3-shot) (twitter_gold)	5.11e-13	-12.47
TSS-BC vs MentalBERT (twitter)	4.32e-03	+6.56
TSS-BC vs LLaMA-3-8B (3-shot) (twitter)	0.00e+00	-10.64

Table 6: Conflict-zone examples (Twitter-gold).

	True	MentalBERT	TSS-BC	Text
E1	0	1	0	What Is Stressing America Out? stress mentalhealth US MentalHealthAwareness
E2	1	0	1	tfw you look at your dissertation draft and wonder if you’ve written 15,000 words of nonsense tired
E3	1	1	0	Self medicating with alcohol and antipsychotics Fighting a mind always against me like a broken record in my head Just another reason for my insanity.. MentalHealth poetry MeToo wednesdaythought

Table 6a: Qualitative error-analysis audit summary.

Item	Value	Notes
Shift cases (B correct, A wrong)	682	Extracted across channels
Gold disagreements sampled	200	Cases where human gold and model disagree
Samples analyzed per sheet	200	Fixed audit sample size per qualitative sheet
Workbook export	<i>error_analysis_qualitative.xlsx</i>	Includes ‘All_Channels_Compare’ and ‘Summary’ sheet

Table 7: Leave-One-Domain-Out (LODO) cross-stressor robustness (Dreaddit domains).

Model/Channel	Mean F1 (out)	Std	Min	Max
TSS-A	0.662	0.042	0.628	0.723
TSS-B	0.570	0.045	0.531	0.637
TSS-BC	0.679	0.032	0.653	0.734
TSS-C	0.685	0.015	0.666	0.700
MentalBERT	0.818	0.030	0.772	0.851

Table 8: Unsupervised stylistic phenotyping (Channel C, K-Means $K=3$).

Statistic	Value	95% CI / p	Interpretation
Bootstrap ARI	0.980	[0.961, 0.991] ($n_{\text{boot}}=100$)	High stability of stylistic profiles.
Cluster–dataset AMI	0.281	$p=0.0005$ (perm=2000)	Non-trivial dataset alignment (potential leakage).
Cluster–label AMI	0.084	$p=0.0005$ (perm=2000)	Weak but significant stress-label signal.
Sample size / features	$N=12924$, $d=154$	—	$K=3$ clusters in 154-dim style space. Clustering uses $N=12,924$; DoD uses $N=12,906$ (18-sample gap: zero-length POS sequences after preprocessing, excluded from DoD but retained in clustering via Channel C features).

Table 9: Instance-level channel orthogonality (Spearman ρ).

Summary	Value	Definition	Implication
Mean $ \rho $ across pairs	0.595	Mean abs. Spearman over all pairs/datasets	Overall weak-to-moderate dependence.
Orthogonality ratio	0.095	Fraction flagged orthogonal	Non-trivial subset weakly correlated.
Key pair: A vs B	0.296	Primary ABC decomposition pair	Lexical vs morpho-syntax most weakly correlated.
Effective N	271,026	$12,906 \times 21$ channel-pairs	Large-sample instance-level audit.

Table 10: Efficiency profile (feature count, training time, and sparsity).

Channel	Features (d)	Train (s)	Regularization	l1_ratio
A	500	17.6	L2 (Ridge)	—
B	6	1007.9	ElasticNet (CV)	0.95
C	154	27.5	L2 (Ridge)	—
BC	160	335.9	ElasticNet (CV)	0.95

Table 11: LLaMA-3 regime-sensitive instability (few-shot).

Dataset	Macro-F1	Prec.	Rec.	Interpretation
Dreaddit-test (H)	0.613	0.606	0.984	Over-sensitivity
Twitter-auto (A)	0.522	0.846	0.203	Under-sensitivity
Twitter-gold (H)	0.800	0.780	0.680	Moderate balance

Table 12: Shift-case taxonomy (A errors corrected by B).

Dataset	n_{shift}	A_FP	A_FN	%A_FP
dreaddit_test	82	42	40	51.2
twitter_gold	200	194	6	97.0
twitter (auto)	200	154	46	77.0
reddit_combi	200	25	175	12.5
ALL	682	415	267	60.9

Table 13: Paired channel contrasts (Δ Macro-F1) by dataset.

Dataset	Compare	Δ F1	95% CI	p_{perm}	N
Dreaddit (H)	C – A	+0.1055	[+0.066, +0.145]	0.0001	715
Dreaddit (H)	BC – A	+0.0566	[+0.016, +0.097]	0.0032	715
Tw-gold (H)	C – A	+0.0389	[+0.015, +0.062]	0.0005	2868
Tw-gold (H)	BC – A	+0.0286	[+0.004, +0.052]	0.0131	2868
Tw-auto (A)	C – A	−0.0186	[−0.034, −0.003]	0.9877	6232
Tw-auto (A)	BC – A	−0.0674	[−0.082, −0.052]	1.0000	6232
Red-combi (A)	C – A	+0.0700	[+0.043, +0.098]	0.0001	3109
Red-combi (A)	BC – A	+0.0779	[+0.051, +0.105]	0.0001	3109

Table 14: Extended metric dashboard (per dataset $\times$ key channels).

Dataset	Ch.	Macro-F1	PR-AUC	MCC	Bal. Acc.	Prev.
Dreaddit (H)	A	0.634	0.809	0.324	0.646	0.516
Dreaddit (H)	C	0.739	0.823	0.495	0.740	0.516
Dreaddit (H)	BC	0.690	0.819	0.421	0.697	0.516
Tw-gold (H)	A	0.661	0.540	0.342	0.680	0.338
Tw-gold (H)	C	0.701	0.637	0.422	0.689	0.338
Tw-gold (H)	BC	0.690	0.708	0.438	0.677	0.338
Tw-auto (A)	A	0.522	0.510	0.065	0.531	0.478
Tw-auto (A)	C	0.504	0.656	0.209	0.568	0.478
Tw-auto (A)	BC	0.455	0.658	0.167	0.545	0.478
Red-combi (A)	A	0.611	0.951	0.223	0.608	0.880
Red-combi (A)	C	0.681	0.964	0.371	0.710	0.880
Red-combi (A)	BC	0.689	0.968	0.380	0.702	0.880

Table 15: Per-stressor LODO breakdown (out-of-domain Macro-F1).

Stressor	TSS-A	TSS-C	TSS-BC	MentalBERT
abuse	0.639	0.672	0.653	0.772
anxiety	0.723	0.700	0.734	0.828
financial	0.632	0.690	0.683	0.833
ptsd	0.686	0.695	0.669	0.851
social	0.628	0.666	0.659	0.806
Mean	0.662	0.685	0.679	0.818
Std	0.042	0.015	0.032	0.030

Table 16: SHAP lexical shortcut concentration (Channel A, ALL datasets).

Mode	<i>n</i> cases	Top-10 conc. mean	95% CI
lexical_wins	40	0.333	[0.308, 0.355]

Top-10 SHAP concentration measures the fraction of total absolute SHAP value captured by the 10 highest-impact lexical features in Channel A when lexical predictions outperform structural ones. A concentration of 33.3% across 500 features indicates moderate shortcut reliance on a small lexical subset.